Heating

Heating systems are one of the main consumers of energy in most homes. Depending on location and available utilities, many different devices—from space heaters to entire central heating systems—are used to warm homes.

3 Water is heated
Heat is transferred to cold water running through pipes looping around the heat exchanger.

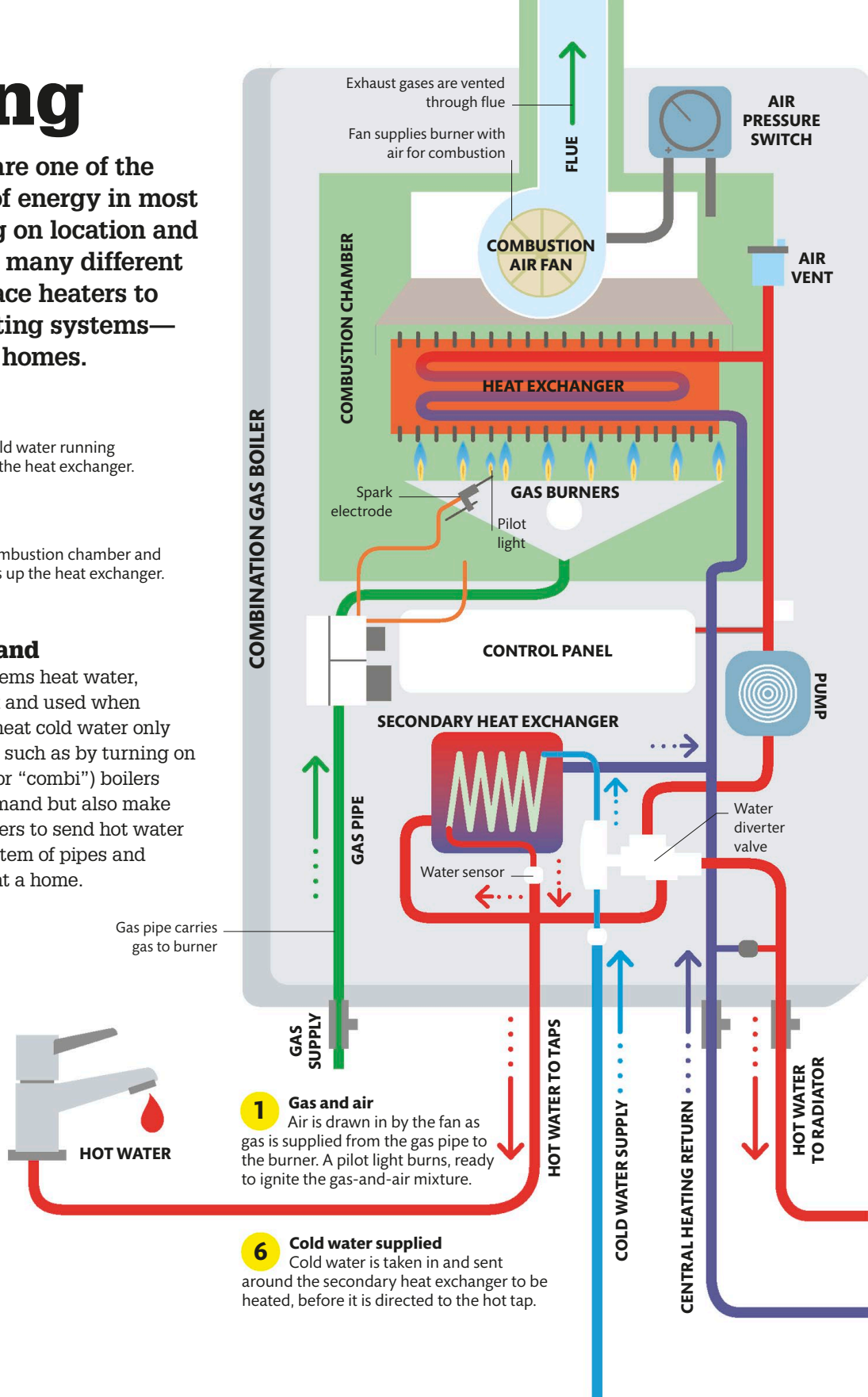
2 Combustion
Gas and air enter the combustion chamber and are ignited. Their burning heats up the heat exchanger.

Hot water on demand

Some home heating systems heat water, which is stored in a tank and used when needed. Other systems heat cold water only when a user demands it, such as by turning on a hot tap. Combination (or "combi") boilers provide hot water on demand but also make use of two heat exchangers to send hot water around a closed-loop system of pipes and radiators to centrally heat a home.

7 Hot water reaches tap
Hot water flows from the tap. When the tap is turned off, the diverter valve closes to allow the central heating to continue.

5 Hot water tap turned on
Turning on a hot water tap makes the boiler's water diverter valve reroute some hot water to the secondary heat exchanger.



1 Gas and air
Air is drawn in by the fan as gas is supplied from the gas pipe to the burner. A pilot light burns, ready to ignite the gas-and-air mixture.

6 Cold water supplied
Cold water is taken in and sent around the secondary heat exchanger to be heated, before it is directed to the hot tap.